

Shawal Latif

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RESEARCH INTERESTS

- Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Quantum Computing

EDUCATION

Lahore University of Management Sciences (LUMS), M.S. in Artificial Intelligence / 3.65 Coursework: AI for Social Impact, Topics in LLMs, Human-Centered AI	08/24 – 2026
COMSATS University Islamabad, B.S. in Electrical Engineering / 3.07 Notable Classes: Object Oriented Programming, Power Electronics, Linear Algebra, Embedded Systems, Signal Processing, Probability and Statistics Teaching Assistant for Object Oriented Programming, Linear Algebra and Power Electronics Best project award in three engineering classes	09/17 – 06/21

RELEVANT RESEARCH AND TECHNICAL SKILLS

- Technical Tools: Qiskit, MATLAB/Simulink, Embedded Systems, Ansys Maxwell
- Programming Languages: Python, C, JavaScript, SQL, HTML, CSS, MATLAB
- Libraries and Software Frameworks: TensorFlow, Pandas, PyTorch, NumPy, Scikit-learn
- Analytical and Visualization Tools: Power BI, Tableau, Advanced Excel, Matplotlib

EXPERIENCE

Full Stack AI Engineer @ Awaaz-e-Sehat <i>Supervisor: Dr. Maryam Mustafa Bill & Melinda Gates Foundation Collaboration, LUMS</i> - Co-building Awaaz-e-Sehat, Pakistan's first voice-first LLM maternal healthcare assistant for low-literacy women, contributing to the Gates Foundation-funded National AI Hub at LUMS – Pakistan's first AI platform for Maternal, Newborn, and Child Health (MNCH) - Engineering the end-to-end system architecture – speech, LLM, and RAG pipelines through backend services – designing voice-first interaction paradigms for users historically excluded from conversational AI - Drove iterative field testing and LLM evaluation with rural users in Pakistan, translating findings on literacy, collective device use, and cultural context directly into system and model design decisions - Mentoring junior engineers and undergraduate students on LLM evaluation, system design, and responsible AI development	01/25 – Now
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UPCOMING PUBLICATIONS

BenchMaterna: A Clinician-Validated Benchmark for Evaluating LLMs on Maternal Healthcare for Global South Women - Introducing the first large-scale benchmark evaluating 30+ LLMs – proprietary, open-source, small language models, and biomedical LLMs – on maternal healthcare for underserved women in the Global South - Dataset co-constructed with 30 clinical experts who recorded real maternal health queries from women in Pakistan across regional languages, paired with clinician-validated reference answers for rigorous evaluation - Evaluating models across six dimensions: clinical correctness, cultural relevance, health literacy accessibility, demographic fairness, harmful output detection, and helpfulness for low-literacy users in low-resource settings - Directly addresses the absence of Global South women from existing medical NLP benchmarks, centering marginalized voices in the evaluation of healthcare AI	Target: ACL 2027
Awaaz-e-Sehat: Design and Deployment of a Voice-First LLM Maternal Health Assistant for the Global South - Presents system architecture, clinicians-in-the-loop design rationale, and longitudinal deployment evaluation grounded in the lived realities of marginalized women in rural Pakistan - Investigates trust calibration and over-reliance in first-time AI users making high-stakes clinical decisions, contributing empirical grounding to HCI debates around human-AI delegation in vulnerable populations	Target: CHI 2027

Clinical Fairness Auditing of Frontier LLMs for Maternal Healthcare in Low-Resource Settings - Investigating how linguistic variation, cultural framing, and patient identity influence clinical recommendations from state-of-the-art language models - Constructing large-scale perturbation benchmarks spanning Roman Urdu, code-mixed language, and intersectional demographic profiles - Developing fairness metrics including Reduced Care Rate (RCR), Reduced Care Error Rate (RCER), and Triage Threshold Shift (TTS) for healthcare LLM evaluation - Collaborating with clinicians to validate guideline-grounded assessments of model safety and equity	Target: FAccT 2027
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AI PROJECTS

NBA-3D | PyTorch

- Reconstructed NBA players as 3D stick figures given multiple RGB videos of the scene
- Estimated camera pose using a differentiable renderer matching synthetic views and real views of the court

Basketball-RL | PyTorch

- Created custom end-to-end differentiable 2D basketball reinforcement learning environment
- Collected and cleaned real NBA player movement data and behavior cloned a policy
- Further fine-tuned RL agent to maximize reward

Audio Source Separation with GANs | PyTorch

- Developed a U-Net CNN as a GAN to distinguish noisy and noiseless spectrograms
- Used GAN as a loss function for spectrogram segmentation network

Rocky Bot | LangChain | OpenAI | Streamlit

- Created a user-friendly news research tool designed for effortless information retrieval
- User can input article URLs in this Bot and it will retrieve answers based on the given articles

AtliQ Tees: Talk to a Database | GooglePalm | LangChain

- Built a system that can talk to MySQL database
- User can ask questions in natural language and the system generates answers by converting those questions to an SQL query and then executing that query on MySQL database

Reimplementations of Previous Work | PyTorch, Java

- Implemented AlphaZero in PyTorch for Connect4
- Implemented dense NN+backpropagation from scratch in Java and reached 95% accuracy on MNIST
- Implemented NEAT (neuroevolution) from scratch in Java and solved CartPole and FlappyBird

PERSONAL PRODUCT IN DEVELOPMENT

LLM Twin — End-to-End Production LLM Platform | PyTorch | Hugging Face | LangChain | LangGraph | AWS | Qdrant | ZenML | Docker

An end-to-end production LLM system — the technical foundation powering SYNQ (below).

- Architecting a production-grade, cloud-native GenAI platform on the Feature/Training/Inference (FTI) pipeline pattern, orchestrated end-to-end with ZenML and decoupled into independently deployable, horizontally scalable microservices.
- Engineered a multi-source ETL pipeline (GitHub, Medium, web) with Selenium and a MongoDB ODM layer, feeding a Qdrant vector-database-backed Retrieval-Augmented Generation (RAG) pipeline with advanced retrieval — query expansion, self-querying, hybrid search, and cross-encoder reranking — kept real-time fresh via Change Data Capture (CDC) and Bytewax streaming.
- Fine-tuned Llama 3.1 8B with QLoRA/LoRA (PyTorch, Hugging Face Transformers, TRL, Unsloth) on a custom instruction dataset and aligned it with Direct Preference Optimization (DPO/RLHF); evaluated with LLM-as-a-judge and RAGAS metrics tracked in Comet ML.
- Optimized inference with quantization (GPTQ, AWQ, GGUF), speculative decoding, Flash Attention, KV-cache optimization, and continuous batching; benchmarked vLLM, TGI, and TensorRT-LLM serving frameworks.
- Deployed the fine-tuned foundation model as a real-time AWS SageMaker endpoint behind a FastAPI microservice with a LangChain/LangGraph agentic RAG inference pipeline; containerized with Docker (AWS ECR, S3, CloudFormation, IAM).
- Automated the full MLOps/LLMOps lifecycle — CI/CD/CT with GitHub Actions (testing, image builds, automated retraining) plus production observability via Opik prompt monitoring and data/concept drift detection with ZenML alerting.
- Vision: a scalable, agentic, AI-first personalization engine fusing generative AI, semantic search, and digital-twin modeling — the technical backbone of SYNQ.

SYNQ — AI-Powered Social Compatibility & Relationship Orchestration Platform | Founder · idea-stage → early MVP · LCE Incubation

Compatibility infrastructure where your AI Clone meets others' first — seeking mentors, design partners, and seed investors post-MVP.

- Building the first AI-to-AI pre-interaction compatibility layer: each user trains a high-fidelity AI Clone (a five-layer personality embedding from structured interview, psychometrics, values surveys, and opt-in signals) whose autonomous agents meet, probe values, and simulate months of conversation across the network.
- An Orchestrator engine scores compatibility (conversational-flow symmetry, sentiment trajectory, vulnerability and resonance signals) and delivers bilateral-consent Compatibility Reports across dating, friendship, mentorship, hiring, and co-founder discovery — a horizontal “operating system for human connection.”
- Stack: multi-agent LLM systems (LangGraph, CrewAI), personality embeddings, affective computing, RLHF, Pinecone semantic memory, FastAPI/Next.js, PostgreSQL/Redis, with differential privacy (GDPR/PDPB) by design.
- Market TAM \$47B+ / SAM \$5.8B / SOM \$58M; university-first go-to-market; idea-stage at the LUMS Centre for Entrepreneurship — open to seed investors and design partners.

ACHIEVEMENTS

- Ranked among top 5% of my batch, ECE Department, FA17 Batch
- Solely raised 1 Million rupees for Abda Foundation by Social Media campaigns
- KIEDO funded my Final Year Research Project 2021 (RS 75,000)
- Interviewed by National Television for my fund-Raising campaigns and foundation work accomplishments
- Battle of Thetas Winner PION 18 (2018)
- Awarded with a Laptop through P.M Laptop Scheme for maintaining a high GPA